
Overcoming Resolution Limitations of 60 MHz Benchtop NMR Using Artificial Intelligence

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Abstract

Low-field benchtop nuclear magnetic resonance (NMR) spectroscopy provides a portable and cost-effective alternative to high-field instruments; however, its reduced spectral resolution has traditionally limited reliable analysis of complex food matrices. In this study, we demonstrate that these intrinsic limitations can be effectively overcome by combining derivative-based spectral enhancement with artificial intelligence, enabling both robust classification and quantitative analysis using 60 MHz ^1H NMR data. A second-derivative transformation was first applied to benchtop NMR spectra to restore effective resolution and improve feature extraction for machine-learning-based analysis. Roasted coffee was employed as a challenging model system to validate methodological improvement under low-field conditions. Second-derivative-processed spectra achieved high classification accuracies for coffee variety and geographical origin (92.2–100%), comparable to those obtained using high-field (600 MHz) NMR, confirming that derivative preprocessing can recover discriminatory power lost at low magnetic field strength. The framework was subsequently extended to a practically relevant adulteration scenario involving high-value edible oils. Using 60 MHz NMR data alone, both classification and quantitative regression of sesame oil blends were successfully performed. Among the evaluated models, a one-dimensional convolutional neural network (1D-CNN) effectively handled severe signal overlap inherent to low-field spectra, achieving superior regression performance ($R^2 = 0.997$, RMSEP = 1.66) and maintaining prediction errors within $\pm 2\%$ for external test samples. Overall, these results demonstrate that low-field benchtop NMR, when integrated with derivative-based spectral enhancement and explainable artificial intelligence, overcomes resolution-related constraints while substantially enhancing practical applicability. This approach provides a reliable, cost-efficient, and broadly deployable platform for routine food authentication and quantitative adulteration analysis.

1. Introduction

Low-field benchtop nuclear magnetic resonance (NMR) spectroscopy has attracted increasing attention as a portable, cost-effective, and maintenance-friendly alternative to high-field NMR systems. Owing to its cryogen-free operation and compact design, benchtop NMR enables chemical analysis in non-specialized environments and offers clear advantages for routine screening and on-site applications. However, its reduced magnetic field strength inevitably results in broader linewidths and severe spectral overlap, which limit its applicability to complex food matrices and have traditionally confined its use to relatively simple classification tasks.

Food authentication represents a particularly challenging application domain, as economically motivated adulteration and mislabeling frequently involve subtle compositional differences rather than complete substitution. Coffee and high-value edible oils are representative examples of such products[1]. Coffee is one of the most widely traded commodities worldwide and is highly susceptible to fraud related to variety and geographical origin, both of which strongly influence market value[2-5]. Similarly, premium oils such as sesame oil are often adulterated through partial blending with lower-cost oils, making reliable detection and quantification across a wide range of blend ratios essential for regulatory enforcement and consumer protection.

Conventional analytical approaches for food authentication—including chromatographic, spectroscopic, and isotope-based techniques—have demonstrated high sensitivity and specificity but often rely on targeted markers or require sophisticated instrumentation and expert operation. In contrast, ^1H NMR spectroscopy provides inherently quantitative, non-destructive, and non-targeted metabolic fingerprints with minimal sample preparation[6, 7]. When combined with chemometric or machine-learning approaches, high-field ^1H NMR has proven highly effective for both classification and quantitative analysis in food authentication. Nevertheless, the high cost and operational complexity of superconducting NMR systems remain major barriers to widespread implementation.

Recent advances in artificial intelligence (AI) and machine learning have opened new opportunities to extract chemically meaningful information from highly overlapped spectral data[8, 9]. In particular, deep learning models such as one-dimensional convolutional neural networks (1D-CNNs) have demonstrated an exceptional ability to learn hierarchical and multi-scale spectral features, outperforming traditional linear and tree-based methods in heavily convoluted signal environments. These capabilities suggest that AI-driven analysis may compensate for the intrinsic resolution limitations of low-field NMR, thereby expanding its practical applicability.

Based on this premise, the present study is structured in two complementary parts to systematically evaluate and extend the capabilities of low-field NMR for food authentication. Part A focuses on methodological validation using coffee as a model system. Here, a second-derivative transformation is applied to 60 MHz ^1H NMR spectra to enhance spectral resolution, and multiple machine-learning and deep-learning models are evaluated for the classification of coffee variety and geographical origin. By benchmarking low-field results against high-field (600 MHz) NMR data, this part establishes that derivative-based spectral enhancement can effectively restore discriminatory power under low-field conditions[10-12].

Part B extends the validated low-field NMR–AI framework to a practical adulteration scenario involving sesame oil. Unlike Part A, this section relies exclusively on 60 MHz NMR data and addresses both binary classification (pure versus adulterated) and quantitative regression of sesame oil content in blended samples. By integrating explainable AI techniques, including SHAP and Grad-CAM, this part further elucidates how different models extract chemically relevant features from low-field spectra and assesses their robustness using external validation samples.

By unifying spectral enhancement strategies with explainable machine learning and deep learning, this study demonstrates that low-field benchtop NMR can overcome its traditional resolution limitations and serve as a reliable, cost-effective platform for both qualitative and quantitative food authentication. The combined results highlight a practical pathway toward deploying NMR-based AI tools beyond specialized laboratories and into real-world regulatory and industrial settings.

Part A. Methodological Validation Using Coffee Samples

2. Experimental Section

Materials and sample preparation

A total of 140 roasted coffee samples were obtained from online markets, comprising Arabica beans from Brazil, Colombia, Ethiopia, and Indonesia (20 samples each) and Robusta beans from India, Vietnam (20 samples each), Indonesia, and Uganda (10 samples each). These samples represent major coffee-producing countries during 2022–2023. Deuterium oxide (D_2O , 99.9 atom% D) containing 1.0% (w/w) DSS was used as the extraction solvent and chemical shift reference. Coffee samples were ground to $<500\ \mu m$, and 180.0 mg of powder was extracted with D_2O , followed by vortexing and centrifugation. An aliquot of the supernatant (700 μL) was transferred to a 5-mm NMR tube for 1H NMR analysis. Extractions were performed in duplicate or quadruplicate, yielding a total of 320 spectra across eight geographical origins.

1H NMR Data Acquisition

Low-field 1H NMR spectra were acquired using a 60 MHz benchtop spectrometer (Magritek Spinsolve-60), referenced to DSS at 0 ppm. For comparison, high-field 1H NMR spectra were collected on a 600 MHz spectrometer (Varian VNS-600) with water suppression using the PURGE pulse sequence. A total of 320 spectra were obtained for each field strength.

Spectral Processing and Data Preparation

All spectra were phase- and baseline-corrected using MestreNova software. The 0.5–9.5 ppm region was retained, excluding the water region (4.505–5.5 ppm), and binned at 0.005 ppm. To mitigate resolution limitations of benchtop NMR, second-derivative preprocessing was applied to the 60 MHz spectra using the Savitzky–Golay algorithm (10-point window, second-degree polynomial), enhancing spectral resolution and feature extraction while minimizing noise amplification.

3. Results

1H NMR spectra of roasted coffee

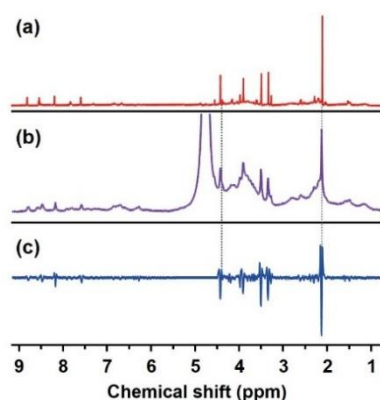


Figure 1. Representative 1H NMR spectra of roasted coffee extracts ($n = 40$ per origin, total 320 samples) acquired using (a) 600 MHz high-field NMR, (b) 60 MHz benchtop NMR, and (c) the second-derivative processed spectrum for 60 MHz data.

Figure 1 shows representative 1H NMR spectra of roasted coffee extracts acquired using 60 MHz benchtop and 600 MHz high-field NMR. The coffee spectra exhibited highly complex resonance patterns, consistent with previous reports [1, 13, 14]. While solvent suppression was applied in high-field NMR, benchtop NMR

required manual exclusion of the water resonance region. Owing to the lower magnetic field strength, 60 MHz spectra showed broadened and highly overlapped signals, particularly in regions containing trigonelline and caffeine, which resulted in baseline distortions. To address these limitations, a second-derivative transformation was applied to the benchtop NMR data. This preprocessing step effectively reduced baseline distortions and improved the separation of overlapping peaks, thereby enhancing the visibility of subtle spectral features[15]. Although derivative processing inherently amplifies noise, Savitzky–Golay (SG) smoothing (10-point window, second-degree polynomial) was applied prior to differentiation to preserve spectral shape while improving the signal-to-noise ratio.

The dominant resonances in the aliphatic region were attributed to caffeine (three N-methyl singlets) and trigonelline, while aromatic signals originated from caffeine, trigonelline, formic acid, and chlorogenic acids, the latter appearing as broad overlapping multiplets.

Discrimination of coffee variety and origin using learning-based approaches

To evaluate classification performance, seven learning-based models were employed, including Random Forest (RF), PCA–LDA, Support Vector Machine (SVM), Artificial Neural Network (ANN), and one-dimensional Convolutional Neural Network (1D-CNN).

Random Forest (RF) is an ensemble-based classifier that aggregates multiple decision trees to improve robustness and reduce overfitting [16]. PCA–LDA combines dimensionality reduction via principal component analysis with linear discriminant analysis to maximize inter-class variance while minimizing intra-class variance [17, 18]. Support Vector Machine (SVM) constructs an optimal maximum-margin hyperplane and is well suited for high-dimensional spectral data [19]. Artificial Neural Network (ANN) learns nonlinear relationships through interconnected layers of neurons and is effective for noisy and overlapping spectral features [20, 21]. One-dimensional Convolutional Neural Network (1D-CNN) applies convolutional filters along the spectral axis, enabling localized feature extraction and improved handling of highly correlated spectral variables [22].

Table 1. Classification accuracies (%) of RF, PCA–LDA, SVM, ANN, and 1D-CNN for roasted coffee using 600 MHz, 60 MHz, and second-derivative processed 60 MHz ¹H NMR spectra (training and test sets).

Spectrum	RF		PCA-LDA		SVM		ANN		1D CNN	
	Training	Test	Training	Test	Training	Test	Training	Test	Training	Test
600 MHz	100	93.8	99.2	92.2	99.2	96.9	98.4	98.4	99.2	100
60 MHz	99.6	81.3	93.4	84.4	98.4	87.5	95.3	84.4	94.9	89.1
2nd Deriv.	99.6	92.2	98.4	92.2	100	93.7	98.8	98.4	98.4	100

The classification accuracies obtained using high-field (600 MHz), low-field (60 MHz), and second-derivative–processed 60 MHz NMR spectra are summarized in Table 1. Using high-field NMR data, all models achieved high test accuracies ranging from 92.2% to 100%, with the 1D-CNN yielding perfect classification performance. RF, PCA–LDA, SVM, and ANN also demonstrated robust results, and training accuracies consistently exceeded 98%, indicating negligible overfitting.

In contrast, classification using raw low-field (60 MHz) spectra resulted in reduced performance. RF and PCA–LDA exhibited lower test accuracies (81.3% and 84.4%, respectively), while SVM and ANN showed moderate performance. Among the evaluated models, the 1D-CNN achieved the highest accuracy under low-field conditions (89.1%), highlighting its superior ability to extract discriminative information from heavily overlapped spectral features.

Application of the second-derivative transformation substantially improved classification performance across all models. Test accuracies increased to 92.2% for RF and PCA–LDA, 93.7% for SVM, and 98.4% and 100% for ANN and 1D-CNN, respectively. Notably, the derivative-processed low-field results closely

matched those obtained using high-field NMR, demonstrating that derivative-based preprocessing effectively compensates for the intrinsic resolution limitations of benchtop systems.

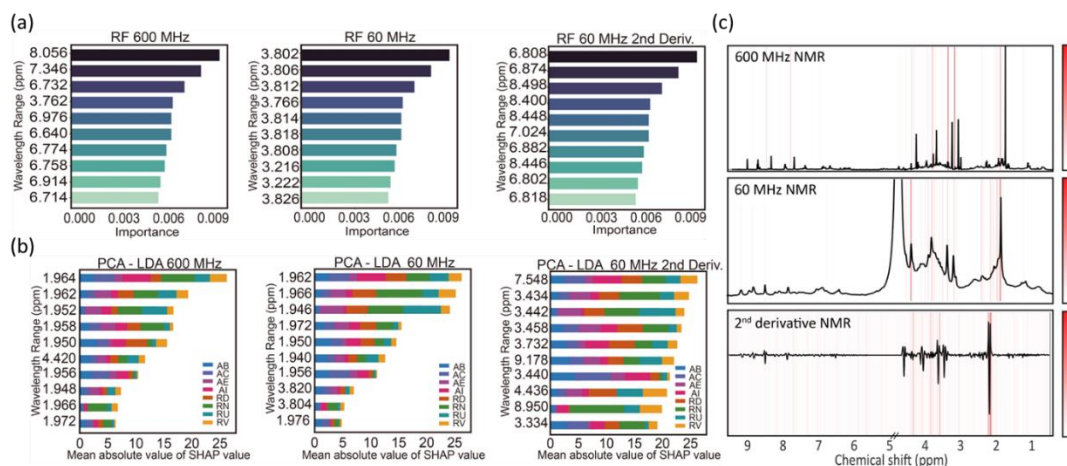


Figure 2. (a) SHAP values for support vector machine (SVM) using 60 MHz benchtop NMR and second-derivative processed 60 MHz spectra, based on the full spectrum. (b) Feature importance for random forest (RF) using 60 MHz and second-derivative processed 60 MHz spectra. (c) Feature attribution heatmaps for one-dimensional convolutional neural network (1D-CNN), highlighting the spectral regions most strongly contributing to classification.

To investigate the spectral features contributing to classification, SHAP and feature-importance analyses were conducted for SVM, RF, and 1D-CNN. As shown in Figure 2, derivative-based preprocessing enabled all models to exploit a broader range of informative spectral regions beyond dominant resonances. This effect was particularly pronounced for ANN and 1D-CNN, indicating that enhanced spectral resolution allowed deep learning models to capture more distributed and chemically meaningful patterns.

Overall, these results demonstrate that second-derivative preprocessing effectively alleviates the intrinsic resolution limitations of low-field NMR, enabling learning-based models—especially deep learning architectures—to achieve classification performance comparable to high-field NMR while maintaining strong generalization capability.

4. Discussion

- **SVM and PCA-LDA:** SHAP analysis indicated that both models, at 600 and 60 MHz, relied primarily on dominant signals such as acetate ($\delta \sim 1.9\text{--}2.0$ ppm) and caffeine ($\delta \sim 3.2\text{--}3.9$ ppm). After applying the second-derivative transformation, the importance expanded to include signals in the aromatic region, particularly trigonelline ($\delta \sim 4.4$, $7.7\text{--}9.1$ ppm), chlorogenic acids ($\delta \sim 6.2\text{--}7.7$ ppm), and formic acid ($\delta \sim 8.4$ ppm), reflecting increased sensitivity to weaker but informative signals.

- **RF and ANN:** RF and ANN exploited a broader set of features than linear models. At 60 MHz, RF depended largely on strong caffeine signals, whereas ANN distributed importance across multiple correlated peaks. After derivative processing, both models increasingly emphasized trigonelline-, chlorogenic-acid-, and formic-acid-related regions.

- **1D-CNN:** The 1D-CNN captured the widest spectral range. While raw 60 MHz data led to reliance on acetate and caffeine, second-derivative preprocessing enabled broader attribution across both dominant and weaker metabolite regions.

Across all models, acetate and caffeine were dominant discriminators. Second-derivative preprocessing reduced dependence on these strong peaks and enabled broader spectral utilization, explaining the improved classification performance under low-field conditions.

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Part B. Authentication and Quantification of Sesame oil

2. Experimental Section

Materials and sample preparation

Three commercial sesame oils and two common adulterants (corn oil and soybean oil; three brands/batches each) were purchased from Korean markets and used to prepare a total of 594 blended oil samples covering the full sesame oil content range (0–100%, v/v, 10% intervals, triplicates). Samples were analyzed by ^1H benchtop NMR after dissolving 100 μL of oil in 600 μL of CDCl_3 (99.8% D) containing 0.03% TMS.

^1H NMR Data Acquisition

Spectra were acquired at 25 $^\circ\text{C}$ on a 60 MHz benchtop spectrometer (Magritek Spinsolve-60) using a 45 $^\circ$ pulse, 3.2 s acquisition time, 10 s relaxation delay, and 64 scans. A total of 594 spectra were obtained for authentication and quantification.

Spectral Processing and Data Preparation

After phase and baseline correction in MNova and referencing to TMS (0.00 ppm), the 0.4–7.0 ppm region was binned at 0.005 ppm resolution (1321 variables) and normalized to total spectral area for AI modeling. The dataset was used for binary classification (100% sesame = pure; 0–90% = adulterated) and regression (0–100% concentration), split into 80/20 training and test sets.

Data augmentation for classification model

To address the severe class imbalance between pure (100%) sesame oil and adulterated samples arising from repurposing regression data for authentication, a SMOTE-type data augmentation strategy was applied by generating synthetic pure samples through linear interpolation of existing pure spectra [23]. This approach models within-class spectral variability and has been shown to improve robustness in spectroscopic classification. In addition, small spectral shifts (1–2 Hz) were introduced to reflect realistic experimental variability [24, 25].

3. Results

Binary classification model for authentication

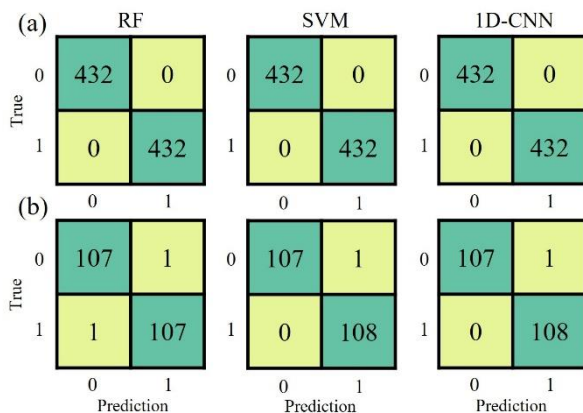


Figure 3. Confusion matrix for binary classification models. (b): result of test set before data augmentation; (b): result of training set; (c): result of test set; 0: pure sesame oil; 1: adulterated oil

Table 2. Accuracy metrics of binary classification models.

Model	Imbalanced		Balanced	
	Train	Test	Train	Test
RF	100%	98.32%	100%	99.07%
SVM	100%	100%	100%	99.54%
1D-CNN	97.89%	96.64%	100%	99.54%

The RF, SVM, and 1D-CNN models all demonstrated excellent classification performance for sesame oil authentication after cross-validation–based optimization, achieving 100% accuracy on the training set and test accuracies exceeding 99%.

The RF model achieved a test accuracy of 99.07%, with one adulterated sample misclassified as pure and one pure sample misclassified as adulterated, where the false-negative case corresponded to a high adulteration level (90%).

The SVM model, optimized with a linear kernel ($C = 0.1$), achieved a test accuracy of 99.54%, with one pure sample incorrectly classified as adulterated.

The 1D-CNN model also achieved a test accuracy of 99.54%, misclassifying one pure sample as adulterated. While the 1D-CNN showed unstable training behavior under imbalanced data conditions, the application of data augmentation resulted in improved training stability and enhanced authentication performance.

The RF, SVM, and 1D-CNN classification models achieved high authentication performance (>99% test accuracy) after cross-validation–based optimization but exhibited distinct responses to data augmentation. RF showed stable performance regardless of augmentation due to its ensemble and bootstrap characteristics, whereas SVM, optimized with a linear kernel, benefited little from interpolation-based augmentation, consistent with its reliance on support vectors near the decision boundary [26]. In contrast, the 1D-CNN classification exhibited unstable training under imbalanced data but showed markedly improved stability and accuracy after data augmentation.

Regression model for quantification

Table 1. Performance metrics of regression models.

Model	Training		Test	
	RMSEC	R ²	RMSEP	R ²
RFR	1.22	0.999	3.07	0.990
SVR	1.55	0.998	2.08	0.996
1D-CNN	0.99	0.999	1.66	0.997

RMSEC, root mean square error of calibration; RMSEP, root mean square error of prediction; RFR, random forest regression; SVR, support vector regression; 1D-CNN, one-dimensional convolutional neural network

Optimized via Bayesian optimization, the 1D-CNN also demonstrated strong regression performance despite using low-field NMR data ($R^2 = 0.999/0.997$ for train/test; RMSEC = 0.99, RMSEP = 1.66), with only a limited number of outliers under a $\pm 2\%$ deviation threshold, indicating superior practical accuracy compared with RFR and SVR.

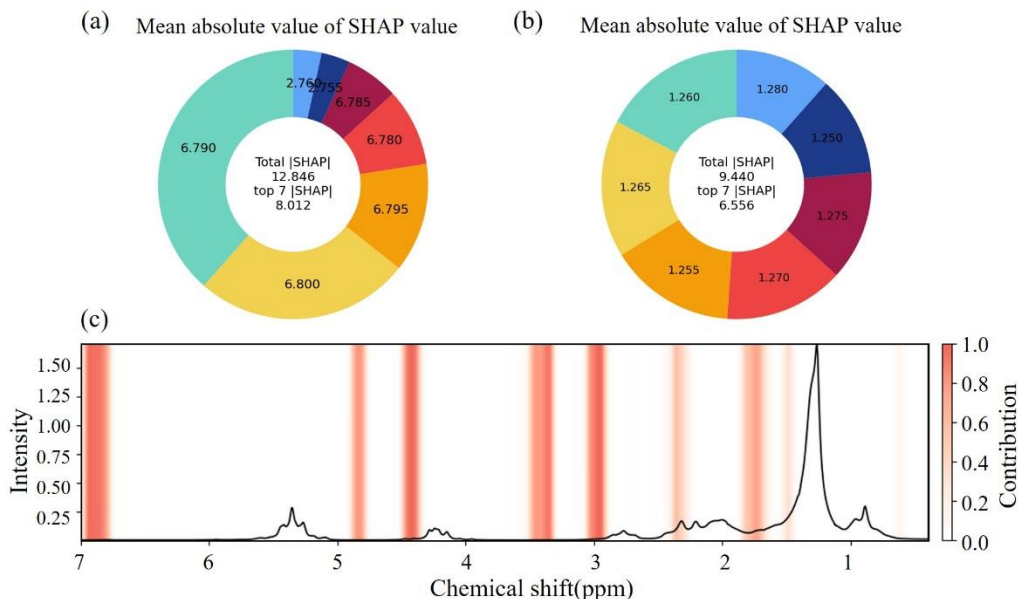


Fig. 5. Feature attribution results for quantification models: SHAP summary plots for (a) RFR and (b) SVR, and (c) Grad-CAM heatmap for 1D-CNN regressor, highlighting significant ^1H NMR spectral regions contributing to model predictions.

To examine how different regression models utilized spectral information for sesame oil quantification, explainable AI (XAI) analyses were applied to RFR, SVR, and 1D-CNN models. Overall, all models relied on chemically meaningful regions of the NMR spectrum but exhibited distinct feature-utilization patterns. The RFR model assigned importance to a limited number of highly discriminative, sesame-specific spectral regions, whereas the SVR model concentrated importance on a small set of non-redundant variables, resulting in sparse saliency patterns [27, 28]. In contrast, the 1D-CNN model exhibited distributed attention across multiple spectral regions, simultaneously incorporating both sesame-specific and lipid-related signals [29]. These differences highlight model-dependent strategies for learning quantitative information from low-field NMR spectra and provide context for the detailed XAI results discussed below.

4. Discussion

- **RFR:** SHAP analysis indicated that importance was concentrated in a limited number of highly discriminative spectral regions, most prominently within the sesame-specific lignan signals at 6.7–6.8 ppm corresponding to sesamol and sesamin. The importance values were distributed over narrow clusters of neighboring bins, reflecting the model's tendency to repeatedly select locally informative features under strong spectral collinearity.

- **SVR:** The model exhibited a much sparser SHAP pattern than RFR. Feature importance was concentrated almost exclusively in the fatty-acid methylene region around 1.2–1.3 ppm, with minimal contribution from other parts of the spectrum. This resulted in peak-centered saliency maps dominated by a small number of non-redundant spectral variables.

- **1D-CNN:** The 1D-CNN model showed a markedly different behavior from the two traditional machine-learning models. Grad-CAM analysis demonstrated that the network learned features distributed across

multiple chemically relevant regions, including sesame-specific bands at 2.9–3.1, 3.2–3.4, and 6.8–6.9 ppm, as well as broader lipid-related regions in the 1–2 ppm range. The Grad-CAM heatmaps indicated that the CNN simultaneously attended to both localized peaks and broader spectral patterns, capturing information across the full spectrum rather than relying on a small subset of variables.

XAI analysis was used to identify which NMR spectral signals contributed most strongly to sesame oil content regression in each model. By highlighting signal-level importance across the spectrum, XAI enabled direct comparison of how different models relied on sesame-specific and lipid-related resonances for quantitative prediction.

5. Conclusion

This study demonstrates that the intrinsic resolution limitations of low-field benchtop NMR spectroscopy can be effectively overcome through the integration of derivative-based spectral enhancement and artificial intelligence, enabling both reliable classification and quantitative analysis of complex food matrices using 60 MHz ^1H NMR data.

In Part A, roasted coffee was employed as a challenging model system to validate the methodological framework under low-field conditions. The application of second-derivative preprocessing substantially improved effective spectral resolution, mitigated baseline distortions, and enabled learning-based models to exploit a broader range of chemically informative signals. As a result, classification accuracies for coffee variety and geographical origin obtained from derivative-processed 60 MHz spectra closely matched those achieved using high-field (600 MHz) NMR, confirming that discriminatory power lost at low magnetic field strength can be recovered without reliance on high-field instrumentation.

In Part B, the validated low-field NMR–AI framework was extended to a practically relevant adulteration scenario involving sesame oil. Using 60 MHz NMR data alone, both authentication and quantitative regression of sesame oil content were achieved with high accuracy. Among the evaluated models, the one-dimensional convolutional neural network (1D-CNN) consistently outperformed traditional machine-learning approaches, demonstrating superior robustness to severe signal overlap and delivering accurate quantitative predictions with low prediction error and strong external validation performance. Explainable AI analyses further confirmed that the models relied on chemically meaningful spectral features, providing transparency and interpretability to the learning process.

Collectively, these results establish that low-field benchtop NMR, when combined with appropriate spectral preprocessing and explainable machine-learning strategies, is not inherently limited to simple classification tasks. Instead, it can serve as a reliable and quantitative analytical platform for food authentication and adulteration analysis. By eliminating the need for high-field instrumentation and minimizing operational complexity, the proposed framework offers a cost-effective, portable, and broadly deployable solution for routine screening, regulatory enforcement, and on-site quality control. This approach provides a practical pathway toward expanding the real-world adoption of NMR-based AI tools beyond specialized laboratories and into industrial and regulatory settings.

Reference

- [1] A. T. Toci, M. V. de Moura Ribeiro, P. R. A. B. de Toledo, N. Boralle, H. R. Pezza, and L. Pezza, "Fingerprint and authenticity roasted coffees by ¹H-NMR: the Brazilian coffee case," *Food Science and Biotechnology*, vol. 27, no. 1, pp. 19-26, 2018.
- [2] G. Campmajo, N. Núñez, and O. Núñez, "The role of liquid chromatography-mass spectrometry in food integrity and authenticity," *Mass Spectrometry—Future Perceptions and Applications; Kamble, GS, Ed.; IntechOpen: London, UK*, pp. 3-20, 2019.
- [3] A. Gonzalvez, S. Armenta, and M. De La Guardia, "Trace-element composition and stable-isotope ratio for discrimination of foods with Protected Designation of Origin," *TrAC Trends in Analytical Chemistry*, vol. 28, no. 11, pp. 1295-1311, 2009.
- [4] S. Kamiloglu, "Authenticity and traceability in beverages," *Food chemistry*, vol. 277, pp. 12-24, 2019.
- [5] J. C. Moore, J. Spink, and M. Lipp, "Development and application of a database of food ingredient fraud and economically motivated adulteration from 1980 to 2010," *Journal of food science*, vol. 77, no. 4, pp. R118-R126, 2012.
- [6] F. Hu, K. Furihata, M. Ito-Ishida, S. Kaminogawa, and M. Tanokura, "Nondestructive observation of bovine milk by NMR spectroscopy: Analysis of existing states of compounds and detection of new compounds," *Journal of agricultural and food chemistry*, vol. 52, no. 16, pp. 4969-4974, 2004.
- [7] F. Hu, K. Furihata, Y. Kato, and M. Tanokura, "Nondestructive quantification of organic compounds in whole milk without pretreatment by two-dimensional NMR spectroscopy," *Journal of agricultural and food chemistry*, vol. 55, no. 11, pp. 4307-4311, 2007.
- [8] F. Lia, B. Vella, M. Zammit Mangion, and C. Farrugia, "Application of ¹H and ¹³C NMR fingerprinting as a tool for the authentication of Maltese extra virgin olive oil," *Foods*, vol. 9, no. 6, p. 689, 2020.
- [9] C. Maione, F. Barbosa Jr, and R. M. Barbosa, "Predicting the botanical and geographical origin of honey with multivariate data analysis and machine learning techniques: A review," *Computers and Electronics in Agriculture*, vol. 157, pp. 436-446, 2019.
- [10] M. Defernez *et al.*, "Low-field ¹H NMR spectroscopy for distinguishing between arabica and robusta ground roast coffees," *Food Chemistry*, vol. 216, pp. 106-113, 2017.
- [11] Y. Gunning *et al.*, "16-O-methylcafestol is present in ground roast Arabica coffees: Implications for authenticity testing," *Food chemistry*, vol. 248, pp. 52-60, 2018.
- [12] K. Speer and P. Mischnick, "16-O-Methylcafestol—a new diterpene in coffee. Discovery and identification," *Zeitschrift für Lebensmittel-Untersuchung und Forschung*, vol. 189, pp. 219-222, 1989.
- [13] R. Consonni, L. R. Cagliani, and C. Cogliati, "NMR based geographical characterization of roasted coffee," *Talanta*, vol. 88, pp. 420-426, 2012.
- [14] V. Gottstein, D. W. Lachenmeier, T. Kuballa, and M. Bunzel, "¹H NMR-based approach to determine the geographical origin and cultivation method of roasted coffee," *Food Chemistry*, vol. 433, p. 137278, 2024.
- [15] L. Rieppo, S. Saarakkala, T. Närhi, H. Helminen, J. Jurvelin, and J. Rieppo, "Application of second derivative spectroscopy for increasing molecular specificity of fourier transform infrared spectroscopic imaging of articular cartilage," *Osteoarthritis and cartilage*, vol. 20, no. 5, pp. 451-459, 2012.
- [16] S. J. Rigatti, "Random forest," *Journal of Insurance Medicine*, vol. 47, no. 1, pp. 31-39, 2017.
- [17] S. Balakrishnama and A. Ganapathiraju, "Linear discriminant analysis-a brief tutorial," *Institute for Signal and information Processing*, vol. 18, no. 1998, pp. 1-8, 1998.
- [18] N. Kumar, A. Bansal, G. Sarma, and R. K. Rawal, "Chemometrics tools used in analytical chemistry: An overview," *Talanta*, vol. 123, pp. 186-199, 2014.
- [19] D. A. Pisner and D. M. Schnyer, "Support vector machine," in *Machine learning*: Elsevier, 2020, pp. 101-121.
- [20] B. Dębska and B. Guzowska-Świder, "Application of artificial neural network in food classification," *Analytica Chimica Acta*, vol. 705, no. 1-2, pp. 283-291, 2011.
- [21] S. Haykin, *Neural networks and learning machines, 3/E*. Pearson Education India, 2009.
- [22] I. L. Jernelv, D. R. Hjelm, Y. Matsuura, and A. Aksnes, "Convolutional neural networks for classification and regression analysis of one-dimensional spectral data," *arXiv preprint arXiv:2005.07530*, 2020.
- [23] S. C. Wong, A. Gatt, V. Stamatescu, and M. D. McDonnell, "Understanding data augmentation for

- classification: when to warp?," in *2016 international conference on digital image computing: techniques and applications (DICTA)*, 2016: IEEE, pp. 1-6.
- [24] E. King, R. Overstreet, J. Nguyen, and D. Ciesielski, "Augmentation of MS/MS Libraries with spectral interpolation for improved identification," *Journal of Chemical Information and Modeling*, vol. 62, no. 16, pp. 3724-3733, 2022.
 - [25] J. Liu, M. Osadchy, L. Ashton, M. Foster, C. J. Solomon, and S. J. Gibson, "Deep convolutional neural networks for Raman spectrum recognition: a unified solution," *Analyst*, vol. 142, no. 21, pp. 4067-4074, 2017.
 - [26] C. J. Burges, "A tutorial on support vector machines for pattern recognition," *Data mining and knowledge discovery*, vol. 2, no. 2, pp. 121-167, 1998.
 - [27] M. L. Guindo *et al.*, "Chemometric approach based on explainable AI for rapid assessment of macronutrients in different organic fertilizers using fusion spectra," *Molecules*, vol. 28, no. 2, p. 799, 2023.
 - [28] H. Zhou *et al.*, "Prediction of maize crude fat content based on improved conditional mutual information maximization and SHAP analysis," *Food Chemistry*, p. 146054, 2025.
 - [29] F. Zhang, Z. Liu, and Y. Wu, "Raman Spectroscopy Combined with a Novel Explainable MSDCNN Networks for Avocado Oil Adulteration Detection," *Journal of Food Composition and Analysis*, p. 107915, 2025.